MATH 243: Discussion Worksheet 10 **Answer Key**

Topics: 15.2 Double Integrals over General Regions; 15.3 Double Integrals in Polar Coordinates; 15.6 Triple Integrals

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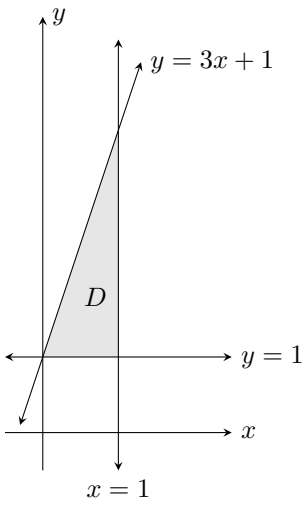
1. Consider the double integral

$$\iint_D f(x,y)\,dA$$

where D is the region bounded by the lines $y = 3x + 1$, $x = 1$, and $y = 1$.

(a) Sketch the region D.

The region D is the closed triangle with vertices $(0,1)$, $(1,1)$, and $(1,4)$.



(b) Set up the double integral as an iterated integral in the order $dy\,dx$ (Type I).

For a fixed x between 0 and 1, a vertical slice of D runs from $y = 1$ to $y = 3x + 1$. Thus

$$D = \{(x,y) \mid 0 \le x \le 1,\ 1 \le y \le 3x+1\}.$$

Therefore,

$$\iint_D f(x,y)\,dA = \int_0^1 \int_1^{3x+1} f(x,y)\,dy\,dx.$$

(c) Set up the double integral as an iterated integral in the order $dx\,dy$ (Type II).

Solving $y = 3x + 1$ for x gives $x = (y-1)/3$. For a fixed y between 1 and 4, a horizontal slice runs from $x = (y-1)/3$ to $x = 1$. Thus

$$D = \left\{(x,y) \;\middle|\; 1 \le y \le 4,\ \frac{y-1}{3} \le x \le 1\right\}$$

and hence

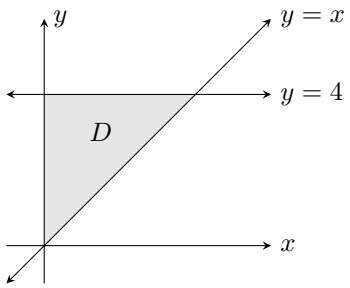
$$\iint_D f(x,y)\,dA = \int_1^4 \int_{(y-1)/3}^1 f(x,y)\,dx\,dy.$$

2. Find the average value of the function $f(x, y) = y^2 e^{xy}$ over the region D bounded by $y = x$, $y = 4$, and $x = 0$.

Recall that

$$f_{\text{avg}} = \frac{1}{A(D)} \iint_D f(x, y)\, dA,$$

where $A(D)$ is the area of D. The region is the triangle with vertices $(0, 0)$, $(0, 4)$, and $(4, 4)$.



The triangle has base 4 and height 4, so $A(D) = \frac{1}{2}(4)(4) = 8$.

As a Type I region, $D = \{(x, y) \mid 0 \le x \le 4,\ x \le y \le 4\}$, giving

$$\iint_D y^2 e^{xy}\, dA = \int_0^4 \int_x^4 y^2 e^{xy}\, dy\, dx.$$

As a Type II region, $D = \{(x, y) \mid 0 \le y \le 4,\ 0 \le x \le y\}$, giving

$$\iint_D y^2 e^{xy}\, dA = \int_0^4 \int_0^y y^2 e^{xy}\, dx\, dy.$$

Integrating with respect to x first avoids integration by parts.

$$\begin{aligned}
\int_0^4 \int_0^y y^2 e^{xy}\, dx\, dy &= \int_0^4 \left[y e^{xy} \right]_{x=0}^{x=y} dy \\
&= \int_0^4 \left(y e^{y^2} - y \right) dy \\
&= \int_0^4 y e^{y^2}\, dy - \int_0^4 y\, dy \\
&= \frac{1}{2} \int_0^{16} e^u\, du - \left[\frac{y^2}{2} \right]_0^4 \\
&= \frac{1}{2} \left[e^u \right]_0^{16} - 8 \\
&= \frac{e^{16} - 17}{2}.
\end{aligned}$$

Therefore, the average value is

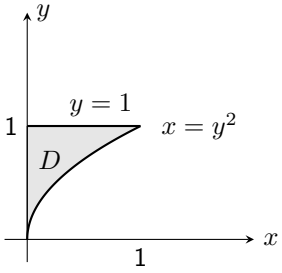
$$f_{\text{avg}} = \frac{1}{8} \cdot \frac{e^{16} - 17}{2} = \frac{e^{16} - 17}{16}.$$

3. Consider the iterated integral

$$\int_0^1 \int_0^{y^2} f(x,y)\,dx\,dy.$$

(a) Sketch the region of integration.

The bounds describe $D = \{(x,y) \mid 0 \le y \le 1,\ 0 \le x \le y^2\}$. For a fixed y, a horizontal slice extends from the y-axis to the parabola $x = y^2$.



(b) Reverse the order of integration.

The x-coordinates in D range from 0 to 1. Because $y \ge 0$, the inequality $x \le y^2$ becomes $y \ge \sqrt{x}$. For a fixed x, a vertical slice therefore extends from $y = \sqrt{x}$ to $y = 1$. Hence

$$\int_0^1 \int_0^{y^2} f(x,y)\,dx\,dy = \int_0^1 \int_{\sqrt{x}}^1 f(x,y)\,dy\,dx.$$

(c) Evaluate the integral in part (b) with the function $f(x,y) = \sqrt{x} + 3y$.

$$\begin{aligned}
\int_0^1 \int_{\sqrt{x}}^1 \left(\sqrt{x} + 3y\right)\,dy\,dx &= \int_0^1 \left[y\sqrt{x} + \frac{3}{2}y^2\right]_{y=\sqrt{x}}^{y=1} dx \\
&= \int_0^1 \left(\sqrt{x} + \frac{3}{2} - x - \frac{3}{2}x\right)\,dx \\
&= \int_0^1 \left(\sqrt{x} + \frac{3}{2} - \frac{5}{2}x\right)\,dx \\
&= \left[\frac{2}{3}x^{3/2} + \frac{3}{2}x - \frac{5}{4}x^2\right]_0^1 \\
&= \frac{2}{3} + \frac{3}{2} - \frac{5}{4} = \frac{11}{12}
\end{aligned}$$

4. Using polar coordinates, find the volume of the solid that lies below the cone

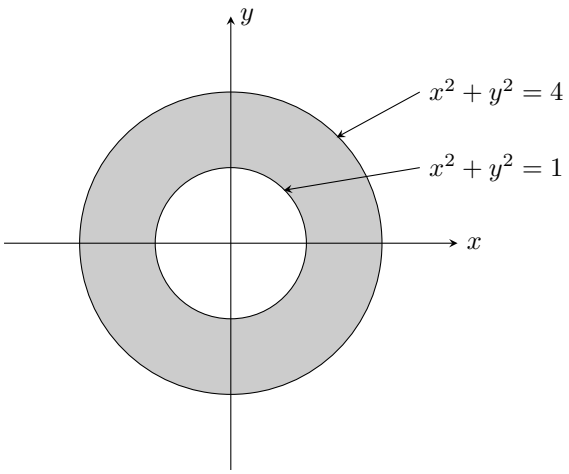
$$z = \sqrt{x^2 + y^2}$$

and above the annulus $1 \le x^2 + y^2 \le 4$ in the xy-plane.

The volume is given by

$$V = \iint_D \sqrt{x^2 + y^2}\, dA$$

where $D = \{(x, y) \mid 1 \le x^2 + y^2 \le 4\}$.



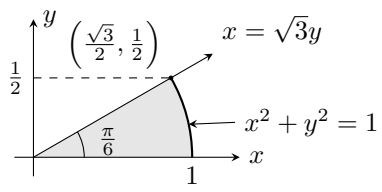
In polar coordinates, the bounds for D are $1 \le r \le 2$ and $0 \le \theta \le 2\pi$. The height of the solid is $\sqrt{x^2 + y^2} = \sqrt{r^2} = r$. Thus,

$$\begin{aligned} V &= \int_0^{2\pi} \int_1^2 r^2\, dr\, d\theta \\ &= \int_0^{2\pi} \left[\frac{r^3}{3}\right]_1^2 d\theta \\ &= \int_0^{2\pi} \frac{7}{3}\, d\theta = \frac{14\pi}{3}. \end{aligned}$$

5. Sketch the region of integration. Then convert to polar coordinates and evaluate

$$\int_0^{1/2} \int_{\sqrt{3}\,y}^{\sqrt{1-y^2}} xy^2\, dx\, dy.$$

The region D is inside the unit circle in the first quadrant and below the line $x = \sqrt{3}y$, or $y = x/\sqrt{3}$. The line meets the circle at $(\sqrt{3}/2, 1/2)$.



On the slanted boundary, $\tan\theta = y/x = 1/\sqrt{3}$, so $\theta = \pi/6$. Thus the polar bounds are $0 \le r \le 1$ and $0 \le \theta \le \pi/6$. Using $x = r\cos\theta$, $y = r\sin\theta$, and $dA = r\,dr\,d\theta$,

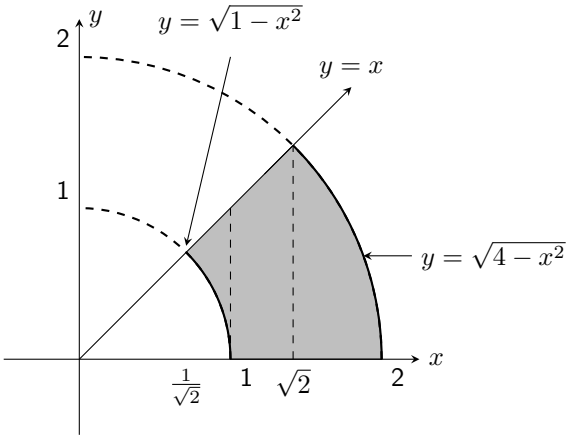
$$\begin{aligned}\iint_D xy^2\,dA &= \int_0^{\pi/6}\int_0^1 (r\cos\theta)(r\sin\theta)^2 r\,dr\,d\theta \\ &= \int_0^{\pi/6}\int_0^1 r^4\cos\theta\sin^2\theta\,dr\,d\theta \\ &= \left[\frac{r^5}{5}\right]_0^1 \int_0^{\pi/6}\cos\theta\sin^2\theta\,d\theta \\ &= \frac{1}{5}\int_0^{\pi/6}\cos\theta\sin^2\theta\,d\theta \\ &= \frac{1}{5}\left[\frac{\sin^3\theta}{3}\right]_0^{\pi/6} = \frac{1}{15}\left(\frac{1}{2}\right)^3 = \frac{1}{120}.\end{aligned}$$

6. The following sum represents $\iint_D xy\,dA$ over a region D:

$$\int_{1/\sqrt{2}}^1\int_{\sqrt{1-x^2}}^x xy\,dy\,dx + \int_1^{\sqrt{2}}\int_0^x xy\,dy\,dx + \int_{\sqrt{2}}^2\int_0^{\sqrt{4-x^2}} xy\,dy\,dx.$$

Sketch D. Express the sum as a single iterated integral in polar coordinates, and evaluate it.

The three integrals describe adjacent vertical portions of the same region: the part of the annulus $1 \le x^2 + y^2 \le 4$ between the positive x-axis and the line $y = x$. The vertical bounds change at $x = 1$, where the inner circle meets the x-axis, and at $x = \sqrt{2}$, where the line $y = x$ meets the outer circle.



In polar coordinates, D is described by $1 \le r \le 2$ and $0 \le \theta \le \pi/4$. Therefore,

$$\begin{aligned}
\iint_D xy\,dA &= \int_0^{\pi/4}\int_1^2 (r\cos\theta)(r\sin\theta) r\,dr\,d\theta \\
&= \int_0^{\pi/4}\int_1^2 r^3\cos\theta\sin\theta\,dr\,d\theta \\
&= \left[\frac{r^4}{4}\right]_1^2 \int_0^{\pi/4} \cos\theta\sin\theta\,d\theta \\
&= \frac{15}{4}\int_0^{\pi/4} \cos\theta\sin\theta\,d\theta \\
&= \frac{15}{4}\left[\frac{\sin^2\theta}{2}\right]_0^{\pi/4} \\
&= \frac{15}{8}\left(\frac{\sqrt{2}}{2}\right)^2 = \frac{15}{16}.
\end{aligned}$$

7. Evaluate each triple integral.

   (a) $\iiint_E (xy + z^2)\,dV$, where $E = [0,2] \times [0,1] \times [0,3]$.

$$\begin{aligned}
\iiint_E (xy+z^2)\,dV &= \int_0^2\int_0^1\int_0^3 (xy+z^2)\,dz\,dy\,dx \\
&= \int_0^2\int_0^1 \left[xyz + \frac{z^3}{3}\right]_{z=0}^{z=3} dy\,dx \\
&= \int_0^2\int_0^1 (3xy+9)\,dy\,dx \\
&= \int_0^2 \left[\frac{3}{2}xy^2 + 9y\right]_{y=0}^{y=1} dx \\
&= \int_0^2 \left(\frac{3}{2}x + 9\right) dx \\
&= \left[\frac{3}{4}x^2 + 9x\right]_0^2 = 3 + 18 = 21.
\end{aligned}$$

   (b) $\iiint_E \dfrac{1}{x^3}\,dV$, where $E = \{(x,y,z) \mid 0 \le y \le 1,\ 0 \le z \le y^2,\ 1 \le x \le z+1\}$.

The given bounds suggest integrating in the order $dx\,dz\,dy$:

$$\begin{aligned}
\iiint_E \frac{1}{x^3}\,dV &= \int_0^1 \int_0^{y^2} \int_1^{z+1} x^{-3}\,dx\,dz\,dy \\
&= \int_0^1 \int_0^{y^2} \left[-\frac{1}{2x^2}\right]_{x=1}^{x=z+1} dz\,dy \\
&= \int_0^1 \int_0^{y^2} \left(\frac{1}{2} - \frac{1}{2(z+1)^2}\right) dz\,dy \\
&= \int_0^1 \left[\frac{z}{2} + \frac{1}{2(z+1)}\right]_{z=0}^{z=y^2} dy \\
&= \int_0^1 \left(\frac{y^2}{2} + \frac{1}{2(1+y^2)} - \frac{1}{2}\right) dy \\
&= \left[\frac{y^3}{6} + \frac{1}{2}\arctan y - \frac{y}{2}\right]_0^1 \\
&= \frac{1}{6} + \frac{\pi}{8} - \frac{1}{2} = \frac{3\pi - 8}{24}.
\end{aligned}$$

8. Set up and evaluate a triple integral to find the volume of the tetrahedron in the first octant bounded by the coordinate planes and the plane passing through $(2, 0, 0)$, $(0, 4, 0)$, and $(0, 0, 12)$.

Using the three intercepts, an equation for the plane is

$$6x + 3y + z = 12.$$

Solving for z gives the upper surface $z = 12 - 6x - 3y$. The solid's projection onto the xy-plane is the triangle described by $0 \le x \le 2$, $0 \le y \le 4 - 2x$. Thus the tetrahedron is given by

$$E = \{(x, y, z) \mid 0 \le x \le 2,\ 0 \le y \le 4 - 2x,\ 0 \le z \le 12 - 6x - 3y\}.$$

Hence,

$$\begin{aligned}
V &= \iiint_E 1\,dV \\
&= \int_0^2 \int_0^{4-2x} \int_0^{12-6x-3y} 1\,dz\,dy\,dx \\
&= \int_0^2 \int_0^{4-2x} (12 - 6x - 3y)\,dy\,dx \\
&= \int_0^2 \left[(12 - 6x)y - \frac{3}{2}y^2\right]_{y=0}^{y=4-2x} dx \\
&= \int_0^2 \left((12 - 6x)(4 - 2x) - \frac{3}{2}(4 - 2x)^2\right) dx \\
&= \int_0^2 (6x^2 - 24x + 24)\,dx \\
&= \left[2x^3 - 12x^2 + 24x\right]_0^2 \\
&= 16 - 48 + 48 = 16.
\end{aligned}$$

SOME USEFUL DEFINITIONS, THEOREMS, AND NOTATION

Double Integrals over General Regions. Suppose that f is continuous on a region D. Assume that the boundary functions below are continuous on their respective intervals.

- A Type I region has the form $D = \{(x, y) \mid a \le x \le b,\ g_1(x) \le y \le g_2(x)\}$. Then

$$\iint_D f(x, y)\, dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y)\, dy\, dx.$$

- A Type II region has the form $D = \{(x, y) \mid c \le y \le d,\ h_1(y) \le x \le h_2(y)\}$. Then

$$\iint_D f(x, y)\, dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y)\, dx\, dy.$$

Double Integrals in Polar Coordinates. Suppose that f is continuous on a region D in the plane, and that D is described in polar coordinates by

$$\alpha \le \theta \le \beta, \qquad h_1(\theta) \le r \le h_2(\theta),$$

where h_1 and h_2 are continuous on $[\alpha, \beta]$, $0 \le h_1(\theta) \le h_2(\theta)$, and $0 < \beta - \alpha \le 2\pi$. Then

$$\iint_D f(x, y)\, dA = \int_\alpha^\beta \int_{h_1(\theta)}^{h_2(\theta)} f(r\cos\theta, r\sin\theta)\, r\, dr\, d\theta.$$

Triple Integrals. Suppose that f is continuous on a solid

$$E = \{(x, y, z) \mid (x, y) \in D,\ u_1(x, y) \le z \le u_2(x, y)\},$$

where D is a Type I or Type II region as above, and u_1, u_2 are continuous on D. Then

$$\iiint_E f(x, y, z)\, dV = \iint_D \left(\int_{u_1(x,y)}^{u_2(x,y)} f(x, y, z)\, dz \right) dA,$$

and we use a Type I or Type II description of D to write this as an iterated integral. Other orders of integration are handled similarly. The volume of E is $V = \iiint_E 1\, dV$.

Suggested Textbook Problems
Section 15.2: 1-50, 55-79 Section 15.3: 1-49 Section 15.6: 1-47, 57, 58